

CS610 Assignment 1 Question 5 (2026)

Your task is to develop a machine learning model to distinguish bot users from genuine users based on a given dataset of Twitter accounts.



Overview Data Discussion Leaderboard Rules Team Submissions

Overview

Your challenge is to build a state-of-the-art fraud detection model that can accurately distinguish between genuine user accounts and sophisticated bot accounts.

Start

18 days ago

Close

7 days to go

Competition Host

Zhao Yiliang



Prizes & Awards

Kudos

Does not award Points or Medals

Participation

47 Entrants

28 Participants

28 Teams

and enhance the quality of its services and to analyze traffic.

Learn more

OK, Got it.

Description



CS610 In-class Kaggle Competition

This competition focuses on an **anomaly detection problem**, where your task is to develop a machine learning model to distinguish **bot users from genuine users** based on a dataset of Twitter accounts.

Through this competition, you will gain hands-on experience in:

- Data preprocessing
- Feature engineering
- Model development
- Model evaluation

Twitter bots are widely used for activities ranging from spam to coordinated disinformation campaigns. Accurately detecting such bots is critical for maintaining the integrity of online platforms. You will be provided with a dataset containing features extracted from Twitter profiles, and your goal is to build a predictive model that can effectively identify bot accounts.

What will the competition be like?

- **Deadline:** 16 May 2026, 11:55 PM (SGT)
- You will be given a **training dataset** and required to generate predictions for a **test dataset**
- Submissions must be in **.csv format**, with a limit of **3 submissions per day**
- Your performance will be evaluated using **AUC (Area Under the ROC Curve)**
- A **public leaderboard** (based on 30% of the test set) will be available during the competition
- A **private leaderboard** (based on the remaining 70%) will determine the **final ranking**
- You may select **up to 2 submissions** for final evaluation on the private leaderboard

How will you be graded?

Total: 55 points

Grading consists of **two components**:

1) Improvement Over Baseline (15 points)

You will receive up to **15 points** if your model demonstrates improvement over the provided baseline. This component evaluates:

- **Working Code**
Your submitted code must run without errors
- **Reproducibility**
Results should be reproducible using your code and provided data
- **Clear Documentation**
Your code should clearly explain:
 - Feature engineering choices
 - Model selection
 - Key design decisions

2) Final Ranking (40 points)

Tags

Roc Auc Score

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40 points are awarded based on your **final ranking on the private leaderboard**.

Your score will be computed using a **scaled (non-linear) transformation**, similar to the approach used for Piazza participation:

$$\text{Final Score} = 40 \times \left(\frac{\text{AUC score}}{\text{Highest AUC score}} \right)^a$$

Where:

- AUC score is your model's private leaderboard score
- Highest AUC score is the best score achieved in the competition
- a is a scaling parameter (e.g., a=5)

Example:

If:

- Your AUC = 0.92
- Highest AUC = 0.95
- a=5

$$\text{Final Score} = 40 \times \left(\frac{0.92}{0.95} \right)^5 \approx 40 \times (0.9684)^5 \approx 40 \times 0.85176 = 34.07$$

Important Instructions

- After joining the competition on Kaggle, **update your Display Name** to your SMU student number (e.g., 01522229)
- **Do not create or use multiple Kaggle accounts**
- This is an **individual competition** — collaboration is not allowed
- Carefully review the **Overview, Data, and Rules** pages before starting

Evaluation



Submissions are evaluated on area under the ROC curve between the predicted probability and the observed target.

Submission File

For each ID in the test set, you must predict a probability for the TARGET variable. The file should contain a header and have the following format:

```
index, target
0,0.9
1,0.1
2,0.5
etc.
```